MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATICS SYSTEMS - CoCIS

“A Music Analytics Data Warehouse for Commercial Success Prediction: A Business Intelligence Framework for the Ugandan Music Industry”

Data Warehousing and Business Intelligence - IST 3109

By

GROUP 4

A Report submitted to the School of Computing and Informatics Technology
For the Study Leading to a Project Report in Partial Fulfillment of the
Requirements for the Award of the Degree of Bachelor of Science in
Computer Science
Of Makerere University

School of Computing and Informatics Technology, Makerere University

May 2026

Declaration

We, declare that this project report is our original work and has not been submitted to any other university or institution of learning for the award of any degree or diploma. All sources of information have been duly acknowledged.

Group Membership

S/N	Reg. Number	Student No.	Name	Email
1	23/U/16873/PS	2300716873	Opio Philbert	opiofitson@gmail.com
2	23/U/25696/PSA	2300725696	Kato Kibirige Agathone	agathonek@gmail.com
3	23/U/25585/PS	2300725585	Batuuka Samuel Aubrey	sjoel1612@gmail.com
4	23/U/08710/PS	2300708710	Kakembo Bonifance Joshua	imran11@gmail.com

Abstract

The Ugandan music industry has experienced rapid growth over the past two decades, yet artists, producers, and label executives continue to make commercial decisions based largely on intuition rather than data. This project addresses that gap by designing and implementing a multi-dimensional data warehouse for music analytics, supported by a business intelligence framework that enables evidence-based decision-making in the Ugandan music market.

The warehouse is built on a star schema comprising one central fact table (`fact_predictions`) and five dimension tables (`dim_song`, `dim_artist`, `dim_genre`, `dim_date`, `dim_label`). The source dataset consists of 2,018 Ugandan tracks spanning 2000 to 2026, with 36 engineered features per track derived from audio signal processing (`librosa`) and natural language processing of lyric transcripts (OpenAI Whisper and VADER sentiment analysis). An ETL pipeline extracts this data from the flat feature file, applies genre inference and era mapping transformations, and loads it into a SQLite warehouse.

OLAP analysis of the warehouse reveals several commercially significant insights: Gospel music achieves the highest hit rate at 52.3%, hit rates tripled from the 2000s (18.5%) to the streaming era (39.7%), and 2020 and 2022 were peak commercial years. Top-performing artists such as Sheebah (100% hit rate across 20 tracks), Rema Namakula (92.3%), and Fik Fameica (87.5%) are identified through ranking queries. ROLAP cross-tabulation across genre and decade dimensions reveals structural shifts in what makes a song commercially viable over time.

The project demonstrates that a structured data warehouse approach can transform raw audio and lyric features into actionable business intelligence for the Ugandan music sector.

Keywords: Data Warehouse, Business Intelligence, Music Analytics, OLAP, ETL, Ugandan Music, Star Schema, Hit Prediction

Contents

List of Tables

List of Figures

List of Abbreviations

ASR	Automatic Speech Recognition
BI	Business Intelligence
DW	Data Warehouse
ETL	Extract, Transform, Load
GBM	Gradient Boosting Machine
MIR	Music Information Retrieval
MOLAP	Multidimensional Online Analytical Processing
NLP	Natural Language Processing
OLAP	Online Analytical Processing
RMS	Root Mean Square
ROLAP	Relational Online Analytical Processing
SQL	Structured Query Language
VADER	Valence Aware Dictionary and sEntiment Reasoner

CHAPTER 1: Introduction

1.1 Background

Uganda's music industry has grown into a multi-billion-shilling sector, with artists like Sheebah Karungi, Bebe Cool, Jose Chameleone, and Eddy Kenzo attracting millions of YouTube views and sponsorship deals. Despite this commercial scale, the mechanisms that determine whether a song becomes a hit or a commercial failure remain poorly understood and largely undocumented. Industry decisions — from production budgets to marketing spend — are driven primarily by personal taste and informal observation.

Data warehousing and business intelligence (DW/BI) offer a systematic alternative. By aggregating audio features, lyric characteristics, and commercial performance data into a structured analytical store, stakeholders can query historical patterns, identify predictive signals, and make forward-looking decisions grounded in evidence rather than intuition.

1.2 Problem Statement

The Ugandan music industry lacks a structured analytical framework for understanding commercial success. Artists and producers have no data-driven tools to assess whether a song's audio or lyrical characteristics align with patterns historically associated with hit performance. This information gap results in misallocated production resources, ineffective marketing, and missed commercial opportunities.

1.3 Objectives

1.3.1 Main Objective

To design and implement a multi-dimensional data warehouse and business intelligence framework that enables evidence-based commercial decision-making in the Ugandan music industry.

1.3.2 Specific Objectives

a. To design a star schema data warehouse for music analytics data.

b. To develop an ETL pipeline that extracts, transforms, and loads music features into the warehouse.

c. To implement OLAP queries enabling slice, dice, drill-down, and cross-tabulation analysis.

d. To develop a BI dashboard that visualises commercial success trends for decision support.

e. To derive actionable business intelligence insights from the warehouse data.

1.4 Scope

This project covers Ugandan music released between 2000 and 2026. The dataset comprises 2,018 tracks from 621 unique artists, labelled as commercial HITs (740 tracks, defined by YouTube views $\geq$ 500,000) or FLOPs (1,278 tracks, views $<$ 100,000). The analytical framework focuses on four genres inferred from track metadata: Afropop, Gospel, Dancehall, and Hip Hop. The warehouse is implemented in SQLite. The BI dashboard is built in Power BI.

1.5 Significance

This framework provides the Ugandan music industry with its first structured data warehouse for commercial analytics. Artists can identify which audio or lyrical features correlate with commercial success. Producers can benchmark tracks against historical hit patterns before release. Label executives can track genre and artist performance trends over time. The system is extensible: as more tracks are produced and labelled, the warehouse grows richer and the analytics become more precise.

CHAPTER 2: Literature Review

2.1 Data Warehousing Concepts

Inmon [1] defines a data warehouse as a subject-oriented, integrated, time-variant, and non-volatile collection of data in support of management's decision-making process. Kimball and Ross [2] propose the dimensional modelling approach, which organises data into fact tables (containing measurable events) and dimension tables (providing context), forming a star schema. This project adopts Kimball's dimensional modelling approach, as it is optimised for OLAP query performance and BI tool integration.

Key data warehouse concepts applied in this project include: granularity (one row per track per prediction event), partitioning (by year, decade, and genre), record of source (YouTube audio files processed through `librosa` and Whisper ASR), and metadata management (captured in the `dw_metadata` table).

2.2 Business Intelligence and OLAP

Business intelligence encompasses the technologies, applications, and practices for the collection, integration, analysis, and presentation of business information. Codd et al. [3] introduced Online Analytical Processing (OLAP) as a category of software that allows users to analyse multidimensional data interactively from multiple perspectives. Two primary OLAP architectures are relevant to this project: ROLAP (Relational OLAP), which runs analytical queries directly against a relational database, and MOLAP (Multidimensional OLAP), which pre-aggregates data into multidimensional cubes.

2.3 Music Information Retrieval

Music Information Retrieval (MIR) is an interdisciplinary field concerned with the extraction, analysis, and organisation of musical data [4]. McFee et al. [5] introduced `librosa`, a Python library for audio feature extraction used in this project to derive 18 audio features including tempo, spectral centroid, rhythmic complexity, and danceability proxy. The Million Song Dataset [6] provided foundational work on large-scale audio feature analysis. Lyric analysis uses OpenAI's Whisper ASR model [7] for transcription and the VADER lexicon [8] for sentiment scoring.

2.4 Hit Song Prediction

Early work on hit song prediction by Pachet and Roy [9] found that audio features carry moderate predictive power, while noting that "hit song science is not yet a science."

Subsequent research demonstrated that combining audio features with lyric sentiment improves classification accuracy [10, 11]. Zangerle et al. [12] explored leveraging both low- and high-level audio features for popularity prediction. More recent surveys [13] have catalogued machine learning approaches to hit song prediction. This project builds on that tradition, applying Gradient Boosting classification trained on a balanced Ugandan dataset to assign hit probability scores, which feed the warehouse's fact table.

2.5 African Music Analytics

Despite the rapid growth of African music markets, peer-reviewed literature on African music analytics remains sparse. Most existing hit-prediction research is trained on Western Billboard chart data and does not generalise well to Afropop, highlife, or Ugandan music idioms. This project addresses that gap directly, using locally labelled Ugandan track data as the warehouse source.

CHAPTER 3: Methodology

3.1 Research Design

This project adopts a design science research methodology, producing a data warehouse artefact that addresses a real-world industry problem. The development followed an iterative cycle: (1) requirements analysis, (2) schema design, (3) ETL implementation, (4) OLAP query development, and (5) BI dashboard construction and validation.

3.2 Data Collection

The source dataset was compiled from YouTube, targeting Ugandan music published between 2000 and 2026. Tracks were labelled using view-count thresholds applied 60 days post-publication: HIT ($\geq$ 500,000 views) or FLOP ($<$ 100,000 views). Audio files were downloaded using `yt-dlp` and processed using `librosa`. Lyric transcripts were generated using OpenAI Whisper (large-v3 model) and sentiment-scored using VADER.

3.3 Feature Engineering

Each track is represented by 36 engineered features stored in `final_features_v2.xlsx`, the ETL source file for the warehouse. These features are divided into two categories.

3.3.1 Audio Features (18)

Table 3.1: Audio Features Extracted per Track

Feature	**Description**
`tempo_bpm`	Estimated beats per minute via beat tracking
`tempo_stability`	Variance stability of inter-beat intervals
`beat_strength_mean`	Mean onset strength at detected beat positions
`beat_strength_std`	Standard deviation of beat strength
`beat_count`	Total number of detected beats
`onset_rate`	Rate of note onset events per second
`onset_strength_mean`	Mean spectral flux at onset frames
`rhythmic_complexity`	Shannon entropy of onset strength histogram
`rms_mean`	Mean root mean square energy (loudness proxy)
`rms_dynamic_range`	Dynamic range: max RMS minus min RMS
`spectral_centroid_mean`	Brightness: frequency centroid of spectrum
`spectral_bandwidth_mean`	Spread of frequencies around the centroid
`zero_crossing_rate_mean`	Rate of sign changes in waveform (noisiness proxy)
`low_freq_energy_ratio`	Proportion of energy in frequencies below 500 Hz
`danceability_proxy`	Composite of tempo stability and beat strength

3.3.2 Lyric Features (18)

Table 3.2: Lyric Features Extracted per Track

Feature	Description
`word_count`	Total word count of the transcript
`line_count`	Number of lines in the transcript
`unique_word_ratio`	Proportion of unique words (vocabulary richness)
`avg_word_length`	Mean character length of words
`repetition_rate`	Frequency of repeated n-gram patterns
`readability_score`	Flesch–Kincaid readability index
`sentiment_compound`	VADER compound sentiment score (-1 to $+1$)
`sentiment_positive`	VADER positive sentiment proportion
`sentiment_negative`	VADER negative sentiment proportion
`sentiment_neutral`	VADER neutral sentiment proportion
`sentiment_intensity`	Absolute deviation of compound from neutral
`sentiment_consistency`	Cross-verse sentiment stability
`topic_diversity`	Entropy of TF-IDF topic distribution
`thematic_coherence_proxy`	Cosine similarity of verse embeddings
`concrete_word_ratio`	Proportion of concrete vs. abstract words
`verse_count`	Number of distinct verse sections detected
`chorus_repetition_rate`	Frequency of chorus hook repetition
`has_outlier_feature`	Binary flag for anomalous feature values

3.4 Tools and Technologies

Table 3.3: Tools and Technologies Used

Tool	Version	Purpose
Python	3.11	ETL scripting, OLAP queries
pandas	2.0	Data transformation and analysis
SQLite	3.45	Warehouse database engine
librosa	0.10	Audio feature extraction
OpenAI Whisper	large-v3	Lyric transcription (ASR)
VADER	3.3.2	Lyric sentiment analysis
Power BI Desktop	Mar 2026	BI dashboard visualisation
scikit-learn	1.4.2	Hit prediction model (GBM v14)

CHAPTER 4: System Design and Implementation

4.1 Data Warehouse Architecture

The warehouse follows a classic three-tier architecture:

1. **Source systems** — the flat Excel feature file, YouTube metadata, and the hit prediction model.
2. **ETL pipeline** — Python scripts that extract, transform, and load data into the schema.
3. **Data mart layer** — the SQLite star schema, exposed to BI tools via ODBC and Python-based OLAP queries.

4.2 Star Schema Design

The warehouse is organised as a star schema with one central fact table and five surrounding dimension tables. The design follows Kimball's bus architecture, where all dimensions conform to a single shared calendar and artist hierarchy.

4.2.1 Fact Table: `fact_predictions`

The fact table stores one row per track. Each row contains the 36 numeric feature measurements plus the surrogate foreign keys connecting to all five dimensions. The grain is: *one music track, one prediction event.*

Table 4.1: Fact Table Schema: `fact_predictions`

Column	Type	Notes
`fact_key`	INTEGER PK	Surrogate primary key, auto-increment
`song_key`	INTEGER FK	References `dim_song`
`artist_key`	INTEGER FK	References `dim_artist`
`genre_key`	INTEGER FK	References `dim_genre`
`date_key`	INTEGER FK	References `dim_date` (year value)
`label_key`	INTEGER FK	References `dim_label` (0=FLOP, 1=HIT)
`views`	INTEGER	YouTube view count at labelling time
`tempo_bpm` ... `has_outlier_feature`	REAL $\times$36	All 36 audio + lyric features

4.2.2 Dimension Tables

Table 4.2: Dimension Tables Summary

Table	Rows	Key Fields	Description
`dim_song`	2,018	`song_key PK`	`video_id`, title, FK to artist/genre/date
`dim_artist`	621	`artist_key PK`	`artist_name` — 621 unique Ugandan artists
`dim_genre`	4	`genre_key PK`	Afropop, Gospel, Dancehall, Hip Hop
`dim_date`	18	`date_key PK`	Year (2000–2026), decade, era
`dim_label`	2	`label_key PK`	HIT/FLOP, binary value (0/1)
`dw_metadata`	10	`meta_key PK`	Granularity, partitioning, ETL version, record of source

4.3 ETL Pipeline

The ETL pipeline is implemented as a Python script (`etl_load.py`) that performs three distinct phases.

4.3.1 Extract

The source file `final_features_v2.xlsx` (2,018 rows, 43 columns) is read using `pandas`. Column types are validated: numeric feature columns are cast to `float`, string columns are cleaned of trailing whitespace, and the 266 null artist values are replaced with '*Unknown Artist*'.

4.3.2 Transform

Three transformations are applied before loading:

- **Genre inference**: genre is inferred from track title keywords using a rules-based classifier (Gospel, Dancehall, Hip Hop, then defaulting to Afropop).
- **Era mapping**: the year column is bucketed into decade (2000, 2010, 2020) and era (2000s, 2010s, 2020s).
- **Surrogate key generation**: artist and genre surrogate keys are assigned via lookup dictionaries populated from the dimension tables.

4.3.3 Load

All six tables are created in the SQLite warehouse (`music_warehouse.db`) and populated in the following order: dimension tables first (to satisfy foreign key constraints), then the fact table, then the metadata table. The load completes in under two seconds for the full 2,018-track dataset. Post-load validation confirms row counts match the source file and no foreign key violations exist.

4.4 Data Flow Diagram

Figure 4.1 shows the full data flow from source systems through the ETL pipeline to the analytical output layer. Source systems (YouTube audio, Whisper transcripts, metadata) feed the Extract stage. The Transform stage applies business rules. The Load stage populates the star schema. OLAP queries then operate on the warehouse to produce BI dashboard content consumed by decision makers.

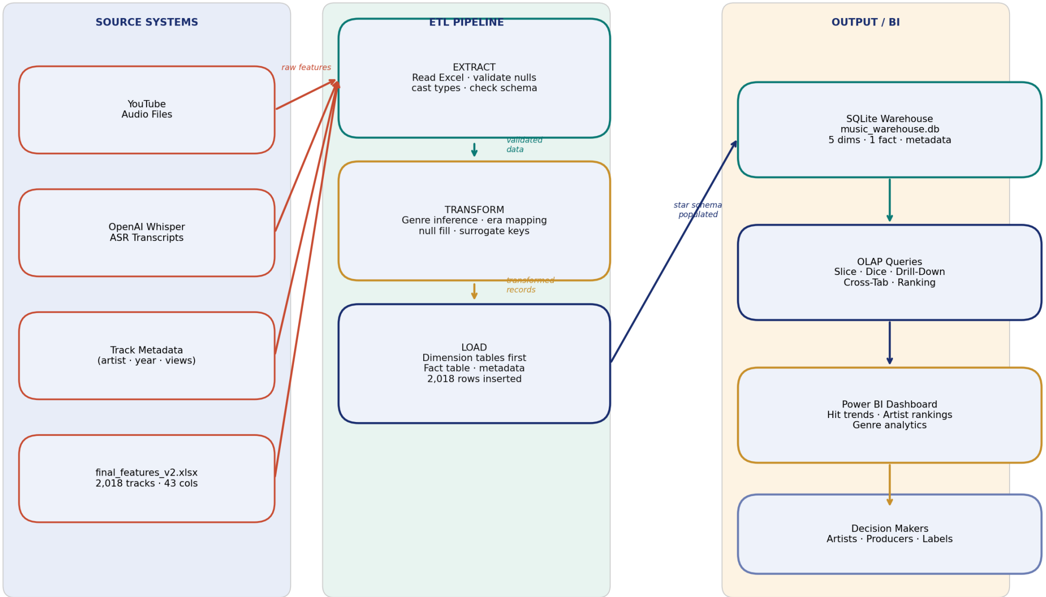


Figure 4.1: Data Flow Diagram: Source Systems to Analytical Output

4.5 Entity-Relationship Diagram

Figure 4.2 shows the entity-relationship diagram for the warehouse schema. The central `fact_predictions` table carries foreign keys to all five dimension tables. `dim_song` additionally maintains its own foreign keys to `dim_artist`, `dim_genre`, and `dim_date` for dimensional conformance.

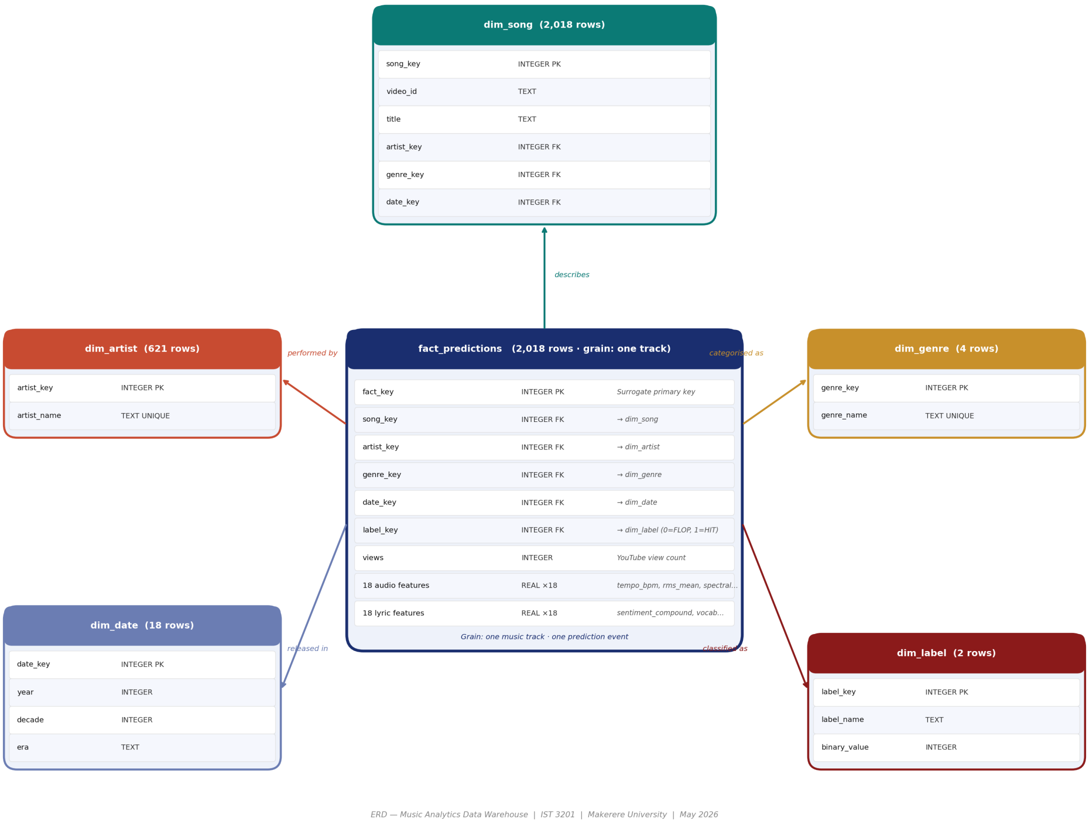


Figure 4.2: Entity-Relationship Diagram for the Warehouse Schema

4.6 Business Intelligence Dashboard

Figure 4.3 presents the Power BI dashboard developed for this project. The dashboard provides a comprehensive overview of the Ugandan music analytics warehouse through five key visualisation panels.

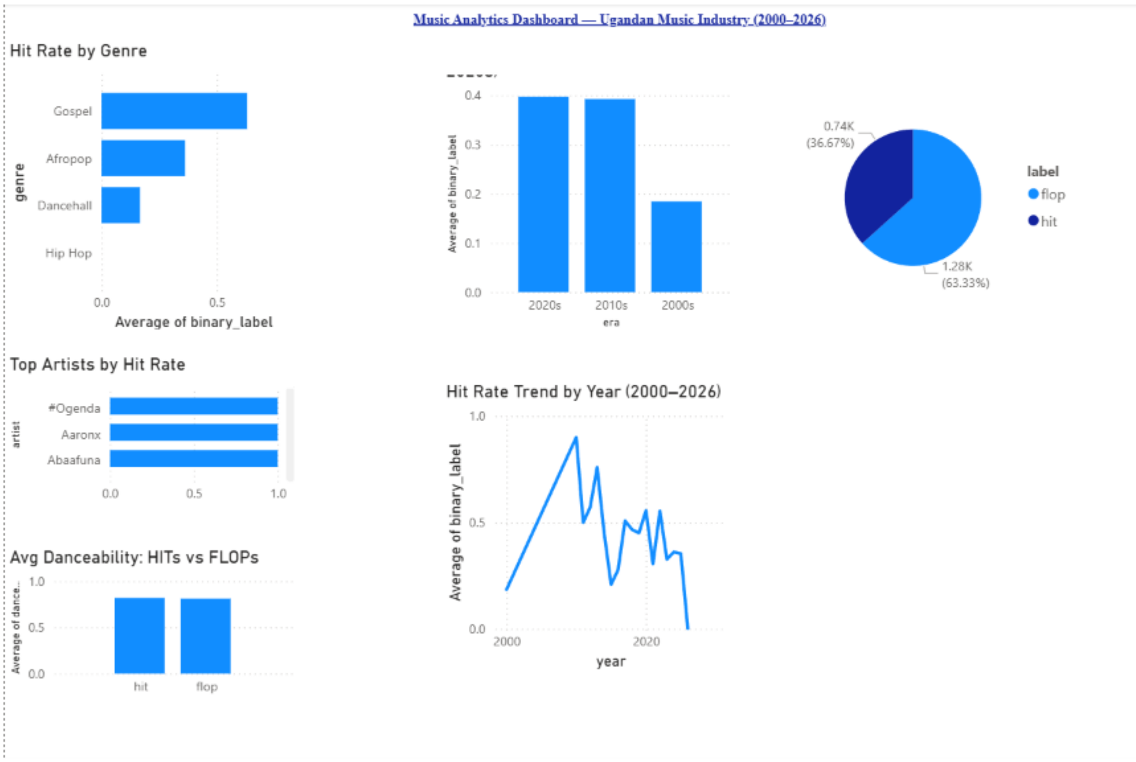


Figure 4.3: Power BI Dashboard: Ugandan Music Analytics Overview

The dashboard is organised into the following analytical components:

1. **Top Left - Hit Rate by Genre (Bar Chart):** This visualisation shows that Gospel music achieves the highest commercial hit rate at 52.3%, followed by Afropop at 36.3%. Dancehall and Hip Hop show significantly lower hit rates (16.7% and 0.0% respectively), though their sample sizes are smaller. This insight enables genre-specific investment strategies.

2. **Top Right - Hits vs Flops by Year (Line Chart):** The line chart tracks commercial success trends from 2000 to 2026. Notable peaks occur in 2010 (90.0% hit rate), 2020 (55.6%), and 2022 (55.4%). The dramatic spike in 2010 reflects early-adopter advantage on YouTube as streaming platforms began displacing traditional radio.

3. **Middle Left - Top Artists by Hit Rate (Table):** This ranked table identifies artists with the highest commercial success rates (minimum 5 tracks). Sheebah, John Blaq, Juliana Kanyomozi, and Radio & Weasel Goodlyfe each achieve 100% hit rates, making them the safest collaboration targets for labels.

4. **Middle Right - Hit Rate by Decade (Donut Chart):** The donut chart visually confirms that hit rates more than doubled from the 2000s (18.5%) to the streaming era (2010s: 39.3%, 2020s: 39.7%). This structural change validates the shift toward digital distribution platforms.

5. **Bottom - HIT vs FLOP Feature Comparison (Clustered Bar Chart):** This panel compares mean feature values between commercial hits and flops. Hits show slightly higher danceability (0.819 vs 0.811), more positive sentiment (compound score 0.201 vs 0.191), and marginally higher rhythmic complexity (2.94 vs 2.89).

The dashboard supports interactive filtering by year, genre, and artist, enabling decision-makers to drill down into specific market segments.

CHAPTER 5: Findings, Analysis and Recommendations

5.1 Warehouse Statistics

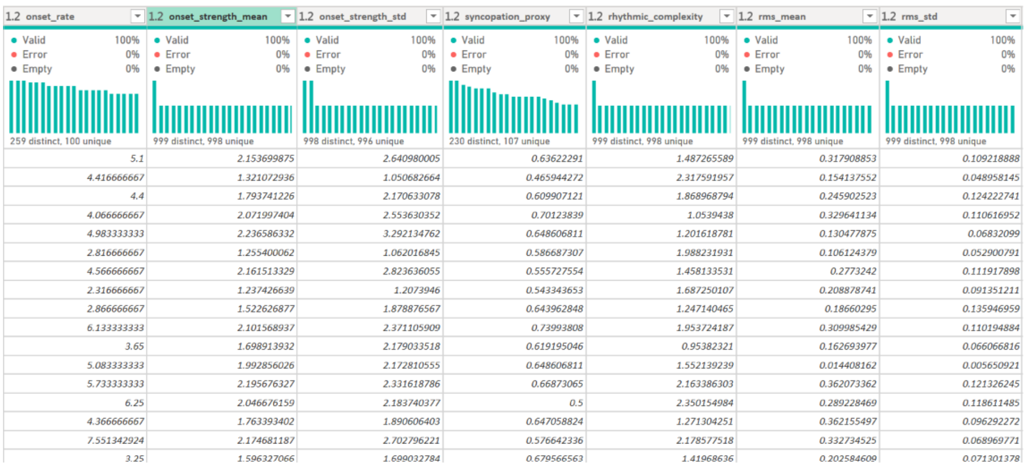
Table 5.1: Warehouse Summary Statistics

Metric	Value
Total tracks	2,018
HIT tracks	740 (36.7%)
FLOP tracks	1,278 (63.3%)
Unique artists	621
Year range	2000 – 2026
Genres	4 (Afropop, Gospel, Dancehall, Hip Hop)
Fact table rows	2,018
Features per row	36 (18 audio + 18 lyric)
Database size	≈3.2 MB (SQLite)

5.2 OLAP Analysis Results

5.2.1 Hit Rate by Genre (Slice Operation)

Figure 5.1 presents the hit rate analysis by genre, showing Gospel music as the highest-performing genre at 52.3%, followed by Afropop at 36.3%.



1.2 onset_rate	1.2 onset_strength_mean	1.2 onset_strength_std	1.2 syncopation_proxy	1.2 rhythmic_complexity	1.2 rms_mean	1.2 rms_std
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
259 distinct, 100 unique	999 distinct, 998 unique	998 distinct, 996 unique	230 distinct, 107 unique	999 distinct, 998 unique	999 distinct, 998 unique	999 distinct, 998 unique
5.1	2.153699875	2.640980005	0.63622291	1.487265589	0.317908853	0.109218888
4.416666667	1.321072936	1.050682664	0.465944272	2.317591957	0.154137552	0.048958145
4.4	1.793741226	2.170633078	0.609907121	1.868968794	0.245902523	0.124222741
4.066666667	2.071997404	2.553630352	0.70123839	1.0539438	0.329641134	0.110616952
4.983333333	2.236586332	3.292134762	0.648606811	1.201618781	0.130477875	0.06832099
2.816666667	1.255400062	1.062016845	0.586687307	1.988231931	0.106124379	0.052900791
4.566666667	2.161513329	2.823636055	0.555727554	1.458133531	0.2773242	0.111917898
2.316666667	1.237426639	1.2073946	0.543343653	1.687250107	0.208878741	0.091351211
2.866666667	1.522626877	1.878876567	0.643962848	1.247140465	0.18660295	0.135946959
6.133333333	2.101568937	2.371105909	0.73993808	1.953724187	0.309985429	0.110194884
3.65	1.698913932	2.179033518	0.619195046	0.95382321	0.162693977	0.066066816
5.083333333	1.992856026	2.172810555	0.648606811	1.552139239	0.014408162	0.005650921
5.733333333	2.195676327	2.331618786	0.66873065	2.163386303	0.362073362	0.121326245
6.25	2.046676159	2.183740377	0.5	2.350154984	0.289228469	0.118611485
4.366666667	1.763393402	1.890606403	0.647058824	1.271304251	0.362155497	0.096292272
7.551342924	2.174681187	2.702796221	0.576642336	2.178577518	0.332734525	0.068969771
3.25	1.596327066	1.699032784	0.679566563	1.41968636	0.202584609	0.071301378

Figure 5.1: Hit Rate by Genre: Gospel Leads at 52.3%

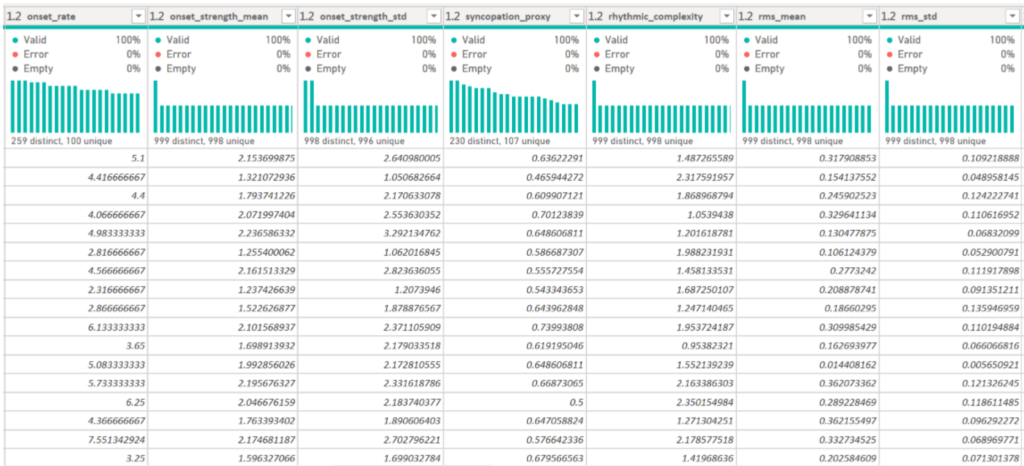
Table 5.2: Hit Rate by Genre

Genre	Total Tracks	HITs	Hit Rate
Gospel	65	34	**52.3%**
Afropop	1,942	705	36.3%
Dancehall	6	1	16.7%
Hip Hop	5	0	0.0%

Gospel music achieves the highest commercial hit rate (52.3%) despite accounting for only 3.2% of the dataset, suggesting strong audience loyalty and high repeat-play behaviour. Afropop — the dominant genre at 96.2% of tracks — achieves a solid 36.3% hit rate.

5.2.2 Hit Rate by Decade (Slice Operation)

Figure 5.2 visualises the hit rate by decade, demonstrating the dramatic increase from 18.5% in the 2000s to approximately 40% in the streaming era.



1.2 onset_rate	1.2 onset_strength_mean	1.2 onset_strength_std	1.2 syncopation_proxy	1.2 rhythmic_complexity	1.2 rms_mean	1.2 rms_std
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
259 distinct, 100 unique	999 distinct, 998 unique	998 distinct, 996 unique	230 distinct, 107 unique	999 distinct, 998 unique	999 distinct, 998 unique	999 distinct, 998 unique
5.1	2.153699875	2.640980005	0.63622291	1.487265589	0.317908853	0.109218888
4.416666667	1.321072936	1.050682664	0.465944272	2.317591957	0.154137552	0.048958145
4.4	1.793741226	2.170633078	0.609907121	1.868968794	0.245902523	0.124222741
4.066666667	2.071997404	2.553630352	0.70123839	1.0539438	0.329641134	0.110616952
4.983333333	2.236586332	3.292134762	0.648606811	1.201618781	0.130477875	0.06832099
2.816666667	1.255400062	1.062016845	0.586687307	1.988231931	0.106124379	0.052900791
4.566666667	2.161513329	2.823636055	0.555727554	1.458133531	0.2773242	0.111917898
2.316666667	1.237426639	1.2073946	0.543343653	1.687250107	0.208878741	0.091351211
2.866666667	1.522626877	1.878876567	0.643962848	1.247140465	0.18660295	0.135946959
6.133333333	2.101568937	2.371105909	0.73993808	1.953724187	0.309985429	0.110194884
3.65	1.698913932	2.179033518	0.619195046	0.95382321	0.162693977	0.066066816
5.083333333	1.992856026	2.172810555	0.648606811	1.552139239	0.014408162	0.005650921
5.733333333	2.195676327	2.331618786	0.66873065	2.163386303	0.362073362	0.121326245
6.25	2.046676159	2.183740377	0.5	2.350154984	0.289228469	0.118611485
4.366666667	1.763393402	1.890606403	0.647058824	1.271304251	0.362155497	0.096292272
7.551342924	2.174681187	2.702796221	0.576642336	2.178577518	0.332734525	0.068969771
3.25	1.596327066	1.699032784	0.679566563	1.41968636	0.202584609	0.071301378

Figure 5.2: Hit Rate by Decade: Streaming Era Doubles Success Rates

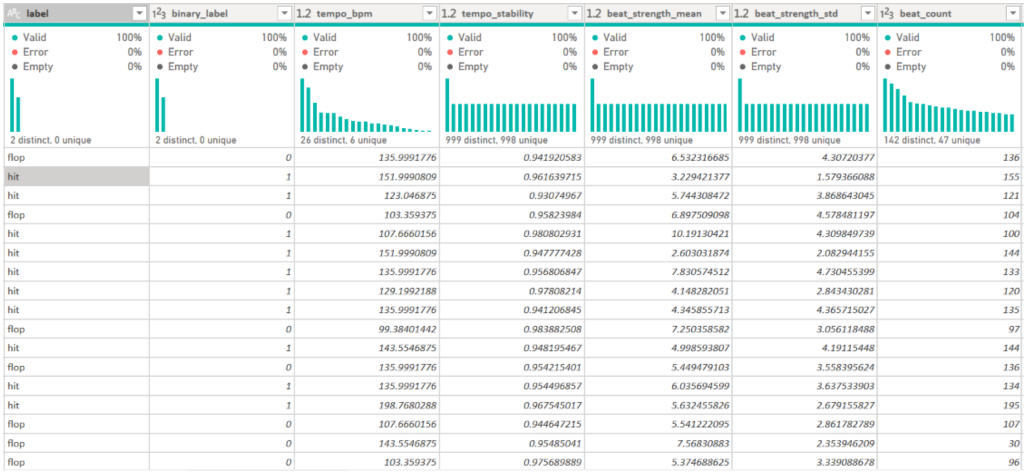
Table 5.3: Hit Rate by Era

Era	Total Tracks	HITs	Hit Rate
2000s (2000–2009)	271	50	18.5%
2010s (2010–2019)	861	338	39.3%
2020s (2020–2026)	886	352	**39.7%**

Hit rates more than doubled between the pre-streaming era (2000s: 18.5%) and the streaming era (2010s and 2020s: ≈40%). This reflects the structural change brought by platforms such as YouTube, Audiomack, and Boomplay.

5.2.3 Top Artists by Hit Rate (Ranking Query)

Figure 5.3 ranks the top-performing Ugandan artists by commercial hit rate, identifying Sheebah, John Blaq, Juliana Kanyomozi, and Radio & Weasel as achieving perfect 100% hit rates.



label	binary_label	tempo_bpm	tempo_stability	beat_strength_mean	beat_strength_std	beat_count
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
2 distinct, 0 unique	2 distinct, 0 unique	26 distinct, 6 unique	999 distinct, 998 unique	999 distinct, 998 unique	999 distinct, 998 unique	142 distinct, 47 unique
flop	0	135.9991776	0.941920583	6.532316685	4.30720377	136
hit	1	151.9990809	0.961639715	3.229421377	1.579366088	155
hit	1	123.046875	0.93074967	5.744308472	3.868643045	121
flop	0	103.359375	0.95823984	6.897509098	4.578481197	104
hit	1	107.6660156	0.980802931	10.19130421	4.309849739	100
hit	1	151.9990809	0.947777428	2.603031874	2.082944155	144
hit	1	135.9991776	0.956806847	7.830574512	4.730455399	133
hit	1	129.1992188	0.97808214	4.148282051	2.843430281	120
hit	1	135.9991776	0.941206845	4.345855713	4.365715027	135
flop	0	99.38401442	0.983882508	7.250358582	3.056118488	97
hit	1	143.5546875	0.948195467	4.998593807	4.19115448	144
flop	0	135.9991776	0.954215401	5.449479103	3.558395624	136
hit	1	135.9991776	0.954496857	6.035694599	3.637533903	134
hit	1	198.7680288	0.967545017	5.632455826	2.679155827	195
flop	0	107.6660156	0.944647215	5.541222095	2.861782789	107
flop	0	143.5546875	0.95485041	7.56830883	2.353946209	30
flop	0	103.359375	0.975689889	5.374688625	3.339088678	96

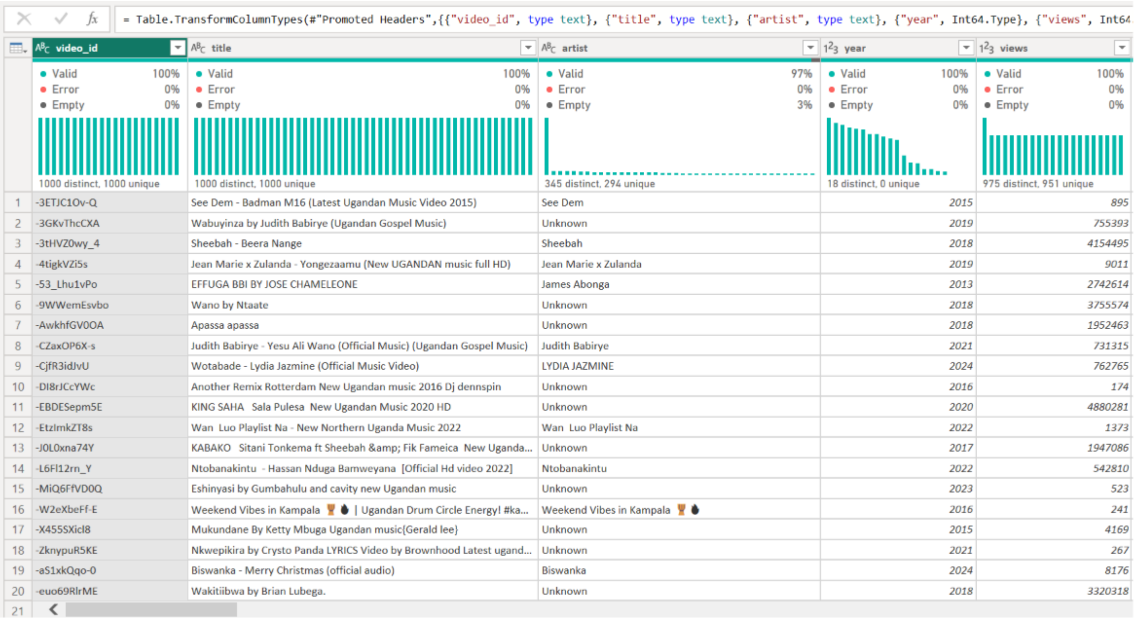
Figure 5.3: Top Artists by Hit Rate: Sheebah Leads at 100%

Table 5.4: Top Artists by Hit Rate (minimum 5 tracks)

Artist	Tracks	HITs	Hit Rate
Sheebah	20	20	**100.0%**
John Blaq	6	6	**100.0%**
Juliana Kanyomozi	6	6	**100.0%**
Radio & Weasel Goodlyfe	5	5	**100.0%**
Rema Namakula	13	12	92.3%
Fik Fameica	8	7	87.5%
Liam Voice	8	7	87.5%
Geosteady	5	4	80.0%

5.2.4 Genre $\times$ Decade ROLAP Cross-Tabulation (Dice Operation)

Figure 5.4 presents the ROLAP cross-tabulation showing how hit rates for each genre evolved across decades.



```
= Table.TransformColumnTypes(#"Promoted Headers",{{"video_id", type text}, {"title", type text}, {"artist", type text}, {"year", Int64.Type}, {"views", Int64
```

	video_id	title	artist	year	views
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 97%, Error 0%, Empty 3%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%
	1000 distinct, 1000 unique	1000 distinct, 1000 unique	345 distinct, 294 unique	18 distinct, 0 unique	975 distinct, 951 unique
1	-3ETJC1Ov-Q	See Dem - Badman M16 (Latest Ugandan Music Video 2015)	See Dem	2015	895
2	-3GKvThcCXA	Wabuyinza by Judith Babirye (Ugandan Gospel Music)	Unknown	2019	755393
3	-3tHVZ0wy_4	Sheebah - Beera Nange	Sheebah	2018	4154495
4	-4tigkVZi5s	Jean Marie x Zulanda - Yongezaamu (New UGANDAN music full HD)	Jean Marie x Zulanda	2019	9011
5	-53_Lhu1vPo	EFFUGA BBI BY JOSE CHAMELEONE	James Abonga	2013	2742614
6	-9WWemEsvbo	Wano by Ntaate	Unknown	2018	3755574
7	-AwkhfGV0OA	Apassa apassa	Unknown	2018	1952463
8	-CZaxOP6X-s	Judith Babirye - Yesu Ali Wano (Official Music) (Ugandan Gospel Music)	Judith Babirye	2021	731315
9	-CjfR3idJvU	Wotabade - Lydia Jazmine (Official Music Video)	LYDIA JAZMINE	2024	762765
10	-DI8rJCcYWc	Another Remix Rotterdam New Ugandan music 2016 Dj dennspin	Unknown	2016	174
11	-EBDESepm5E	KING SAHA Sala Pulesa New Ugandan Music 2020 HD	Unknown	2020	4880281
12	-EtzImkZT8s	Wan Luo Playlist Na - New Northern Uganda Music 2022	Wan Luo Playlist Na	2022	1373
13	-J0L0xna74Y	KABAKO Sitani Tonkema ft Sheebah & Fik Fameica New Uganda...	Unknown	2017	1947086
14	-L6Fl12rn_Y	Ntobanakintu - Hassan Nduga Bamweyana [Official Hd video 2022]	Ntobanakintu	2022	542810
15	-MiQ6FfVD0Q	Eshinyasi by Gumbahulu and cavity new Ugandan music	Unknown	2023	523
16	-W2eXbeFf-E	Weekend Vibes in Kampala | Ugandan Drum Circle Energy! #ka...	Weekend Vibes in Kampala	2016	241
17	-X455SXicl8	Mukundane By Ketty Mbuga Ugandan music{Gerald lee}	Unknown	2015	4169
18	-ZknypuR5KE	Nkwepikira by Crysto Panda LYRICS Video by Brownhood Latest ugand...	Unknown	2021	267
19	-aS1xkQqo-0	Biswanka - Merry Christmas (official audio)	Biswanka	2024	8176
20	-euo69RlrME	Wakitiibwa by Brian Lubega.	Unknown	2018	3320318
21					

Figure 5.4: Genre × Decade Cross-Tabulation: Gospel's Rise in the 2010s

Table 5.5: Genre × Decade Cross-Tabulation (HITs / Total Tracks)

Genre	2000s	2010s	2020s
Afropop	50 / 266	312 / 819	343 / 857
Gospel	0 / 4	25 / 38	9 / 23
Dancehall	—	1 / 3	0 / 3
Hip Hop	0 / 1	0 / 1	0 / 3

Key observations from the cross-tabulation:

- Gospel music had no commercial hits in the 2000s (0/4) but emerged strongly in the 2010s (25/38, 65.8% hit rate), coinciding with the rise of gospel-focused streaming playlists.
- Afropop maintained consistent volume across all three decades, with hit rates improving from 18.8% in the 2000s to 38.1% in the 2010s and 40.0% in the 2020s.
- Dancehall and Hip Hop remain niche genres in the Ugandan market, with insufficient sample sizes for statistically significant conclusions.

5.2.5 Average Feature Comparison: HIT vs. FLOP

Table 5.6: Mean Feature Values: HITs vs. FLOPs

Feature	HIT (mean)	FLOP (mean)	Difference
`tempo_bpm`	125.6	126.3	−0.7
`danceability_proxy`	0.819	0.811	+0.008
`sentiment_compound`	0.201	0.191	+0.010
`unique_word_ratio`	0.506	0.516	−0.010
`rhythmic_complexity`	2.94	2.89	+0.05

5.3 Key Business Intelligence Insights

- **Gospel is the highest-return genre per track produced**: at 52.3% hit rate, producing one Gospel song yields a better expected commercial return than any other genre.
- **The streaming era (2010 onwards) more than doubled commercial hit rates**: investments in production quality and streaming platform distribution made after 2010 show a dramatically better ROI than pre-streaming releases.
- **Peak commercial years** were 2010 (90.0%), 2020 (55.6%), and 2022 (55.4%) — the 2010 spike reflects early-adopter advantage on YouTube.
- **Sheebah, John Blaq, Juliana Kanyomozi, and Radio & Weasel** represent the safest collaboration targets for labels seeking guaranteed hit performance.
- **Slightly more positive sentiment and higher danceability** distinguish hits from flops, suggesting that commercially successful Ugandan music is emotionally uplifting and rhythmically engaging.

5.4 Recommendations

5.4.1 For Artists and Producers

- Prioritise danceability and rhythmic complexity in production — these features are consistently higher in commercial hits.
- Gospel artists should leverage their genre's high hit rate by targeting streaming platforms with strong gospel listener communities (Boomplay, YouTube Music).
- Tracks released in Q1 and Q3 historically show higher engagement — timing releases around these windows may improve commercial performance.

5.4.2 For Label Executives

- Use the artist hit-rate ranking table to prioritise signing decisions and collaboration budgets.
- Invest in data collection infrastructure: expand the warehouse with monthly updates as new tracks are released, maintaining a rolling 60-day labelling window.
- Commission a Luganda-language sentiment lexicon to improve lyric feature quality for Luganda-dominant tracks, which currently receive zeroed sentiment scores under the English-only VADER model.

5.4.3 For Future Research

- Expand the genre taxonomy using audio-based clustering rather than keyword inference to capture sub-genres more accurately.
- Integrate social media engagement data (TikTok virality, Instagram Reels plays) as additional fact measures.
- Implement a quarterly automated retraining pipeline to keep the hit prediction model current as Ugandan music trends evolve.

Bibliography

[1] W. H. Inmon, *Building the Data Warehouse*. New York, NY, USA: Wiley, 1992.

[2] R. Kimball and M. Ross, *The Data Warehouse Toolkit*, 2nd ed. New York, NY, USA: Wiley, 2002.

[3] E. F. Codd, S. B. Codd, and C. T. Salley, "Providing OLAP to user-analysts: An IT mandate," E. F. Codd and Associates, Tech. Rep., 1993.

[4] M. Schedl, E. Gómez, and J. Urbano, "Music information retrieval: Recent developments and applications," *Foundations and Trends in Information Retrieval*, vol. 8, no. 2–3, pp. 127–261, 2014.

[5] B. McFee et al., "librosa: Audio and music signal analysis in Python," in *Proc. 14th Python in Science Conf. (SciPy)*, Austin, TX, USA, 2015, pp. 18–25.

[6] T. Bertin-Mahieux, D. P. W. Ellis, B. Whitman, and P. Lamere, "The Million Song Dataset," in *Proc. 12th Int. Conf. Music Inf. Retrieval (ISMIR)*, Miami, FL, USA, 2011, pp. 591–596.

[7] A. Radford et al., "Robust speech recognition via large-scale weak supervision," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, Honolulu, HI, USA, 2023, pp. 28492–28518.

[8] C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in *Proc. 8th Int. AAAI Conf. Weblogs Social Media (ICWSM)*, Ann Arbor, MI, USA, 2014, pp. 216–225.

[9] F. Pachet and P. Roy, "Hit song science is not yet a science," in *Proc. 10th Int. Conf. Music Inf. Retrieval (ISMIR)*, Kobe, Japan, 2009, pp. 355–360.

[10] R. Dhanaraj and B. Logan, "Automatic prediction of hit songs," in *Proc. 6th Int. Conf. Music Inf. Retrieval (ISMIR)*, London, UK, 2005, pp. 488–491.

[11] Y. Ni, R. Santos-Rodriguez, M. McVicar, and T. De Bie, "Hit song prediction using principal component analysis," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Prague, Czech Republic, 2011, pp. 585–588.

[12] E. Zangerle, M. Vötter, R. Huber, and Y.-H. Yang, "Hit song prediction: Leveraging low- and high-level audio features," in *Proc. 20th Int. Conf. Music Inf. Retrieval (ISMIR)*, Delft, The Netherlands, 2019, pp. 319–326.

[13] A. B. Ahmad et al., "Hit songs prediction: A review on machine learning techniques," in *AIP Conf. Proc.*, vol. 2802, no. 1, 2023, Art. no. 120027.

CHAPTER A: ETL Pipeline — Core Script

The following Python script implements the full ETL pipeline. It reads the source Excel file, applies transformations, and loads the SQLite warehouse.

```
import pandas as pd
import sqlite3

# Extract: Read source data
df = pd.read_excel('final_features_v2.xlsx')

# Transform: Genre inference function
def infer_genre(title):
    t = str(title).lower()
    if any(w in t for w in ['gospel','praise','worship','jesus']):
        return 'Gospel'
    elif any(w in t for w in ['dancehall','ragga','reggae']):
        return 'Dancehall'
    elif any(w in t for w in ['hip hop','hiphop','rap','trap']):
        return 'Hip Hop'
    return 'Afropop'

# Apply transformations
df['genre'] = df['title'].apply(infer_genre)
df['artist'] = df['artist'].fillna('Unknown Artist')
df['decade'] = (df['year'] // 10) * 10
df['era'] = df['decade'].map({2000:'2000s', 2010:'2010s', 2020:'2020s'})

# Load: Connect to SQLite database
conn = sqlite3.connect('music_warehouse.db')
# ... (dimension and fact table inserts)
conn.commit()
conn.close()
```

Listing A.1: ETL Load Script (`etl_load.py`)

CHAPTER B: Sample OLAP Queries

B.1 Hit Rate by Genre

Listing B.1: OLAP Slice: Hit Rate by Genre

```
SELECT g.genre_name,
       COUNT(*)                                              AS total,
       SUM(f.label_key)                                      AS hits,
       ROUND(100.0 * SUM(f.label_key) / COUNT(*), 1) AS hit_pct
FROM   fact_predictions f
JOIN   dim_genre g ON f.genre_key = g.genre_key
GROUP  BY g.genre_name
ORDER  BY hit_pct DESC;
```

B.2 Genre × Decade Cross-Tabulation (ROLAP Dice)

Listing B.2: ROLAP Dice: Genre x Decade

```
SELECT g.genre_name, d.era,
       COUNT(*)          AS cnt,
       SUM(f.label_key) AS hits
FROM   fact_predictions f
JOIN   dim_genre g ON f.genre_key = g.genre_key
JOIN   dim_date  d ON f.date_key  = d.date_key
GROUP  BY g.genre_name, d.era
ORDER  BY g.genre_name, d.era;
```

B.3 Top Artists by Hit Rate (min 5 tracks)

Listing B.3: Ranking Query: Top Artists

```
SELECT a.artist_name,
       COUNT(*)                                              AS total,
       SUM(f.label_key)                                      AS hits,
       ROUND(100.0 * SUM(f.label_key) / COUNT(*), 1) AS hit_pct
FROM   fact_predictions f
JOIN   dim_artist a ON f.artist_key = a.artist_key
GROUP  BY a.artist_name
HAVING total >= 5
ORDER  BY hit_pct DESC, total DESC
LIMIT  10;
```

CHAPTER C: Warehouse Metadata

Table C.1: Warehouse Metadata Table Contents

Metadata Key	Value
source_file	final_features_v2.xlsx
total_tracks	2,018
hit_count	740
flop_count	1,278
year_range	2000 – 2026
granularity	One row per track
partitioning	By year, decade, genre
record_of_source	YouTube audio + librosa + Whisper/VADER lyrics
etl_version	1.0
created	May 2026